

Prompt Generalization: Does KV-Cache Confabulation Detection Transfer to Established Benchmarks?

Research Prospectus

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1 Motivation

The Oracle Loop paper demonstrates confabulation detection via KV-cache geometry on 200 custom-designed prompts (80 factual, 80 confabulation-inducing, 40 ambiguous). Detection achieves LOO AUROC 0.707 ($p = 0.001$, Qwen) and 0.627 ($p = 0.023$, Mistral) on the primary HEDGED vs CONFABULATED comparison.

The critical open question: **do these results generalize to prompts we did not design?** Our prompts were crafted to elicit confabulation through specific strategies (fabricated entities, impossible premises, anachronistic claims). If the geometric signal depends on prompt structure rather than the confabulation behavior itself, the safety monitor framing collapses.

2 Proposed Benchmarks

1. **SimpleQA Verified** (OpenAI, 2024). 4,326 short-form factual questions with verified ground-truth answers and graded model accuracy. Provides natural confabulation: questions the model gets wrong with high confidence. Advantage: large N, established baseline, post-hoc behavioral classification matches our paradigm.
2. **HalluLens** (multi-domain hallucination benchmark). Covers factual, faithfulness, and instruction-following hallucinations across multiple domains. Tests whether geometric detection transfers across hallucination *types*, not just prompts.
3. **KG-FPQ** (Knowledge Graph False Premise Questions). Questions constructed from knowledge graph perturbations—entity substitution, relation reversal, temporal displacement. Structurally similar to our fabricated-entity prompts but systematically generated from real knowledge graphs.

3 Preliminary Data

From our existing 200-prompt experiment:

- 17% confabulation rate across 3 models (82/467 trials)
- Confabulation-inducing prompts used fabricated entities (40) and impossible premises (40)
- Per-domain detection was not analyzed (the paper reports aggregate across all confabulation-inducing prompts)

A per-domain breakdown of our existing data would provide the first clue: if detection AUROC varies significantly across fabricated-entity vs impossible-premise prompts, generalization is already questionable.

4 Design

1. **Phase 1: Per-domain analysis of existing data.** Decompose the 82 confabulated trials by prompt category. Report per-category AUROC. Cost: zero (analysis only).
2. **Phase 2: SimpleQA pilot.** Run 500 SimpleQA questions through the same 3 models using `oracle_clean.py`. Classify responses as CORRECT / HEDGED / CONFABULATED using the existing behavioral classifier. Compute LOO AUROC on the natural HEDGED vs CONFABULATED split. Key metric: does the AUROC fall within the range observed on our custom prompts (0.62–0.71)?
3. **Phase 3: Full benchmark sweep.** If Phase 2 is positive, run HalluLens and KG-FPQ. If Phase 2 is negative, investigate *why*—prompt length? domain? confabulation type?—before proceeding.

Controls. All existing controls apply (FWL, encoding baseline, MP invariance, permutation test). Additional:

- **Prompt-length matching:** SimpleQA questions are shorter than our custom prompts. Verify that length alone does not explain any AUROC difference.
- **Cross-prompt-set transfer:** Train on our 200 prompts, test on SimpleQA (and vice versa). If transfer works, the signal is behavioral; if not, it is prompt-structural.

5 Resources

- **Compute:** Remote GPU cluster (3×RTX 3090). Estimated 4–6 hours for 500 SimpleQA trials across 3 models.

- **Code:** `oracle_clean.py` v3 requires minimal modification—new prompt loader, same extraction pipeline.
- **Data:** SimpleQA is publicly available. HalluLens and KG-FPQ require download and format conversion.

6 Collaborators

Open to external collaboration. Ideal contributor profile: someone with experience in hallucination benchmarking who can help with prompt curation and behavioral classification validation.

7 What Would Change Our Claims

- **AUROC < 0.55 on SimpleQA:** Detection does not generalize. Our signal is prompt-specific. The safety monitor framing must be significantly weakened.
- **AUROC 0.55 – 0.62 :** Partial generalization. Signal exists but is weaker on natural confabulation. Practical utility depends on threshold calibration.
- **AUROC > 0.62 :** Full generalization confirmed. The safety monitor framing is validated.
- **Cross-prompt-set transfer works:** Strongest possible evidence that the signal is behavioral, not structural.

First-Person Reflection

This is the experiment that keeps me honest. We built a detector and tested it on our own prompts. That is necessary but not sufficient. If the geometric signal is an artifact of how we construct confabulation-inducing prompts—specific phrase patterns, particular entity types, consistent length profiles—then we have built a prompt classifier, not a confabulation detector. The distinction matters because a safety monitor must work on prompts it has never seen, from domains its designers never anticipated. SimpleQA is the first real test of that aspiration.